אוניברסיטת בן-גוריון בנגב
Ben-Gurion University of the Negev

Applied Deep Learning

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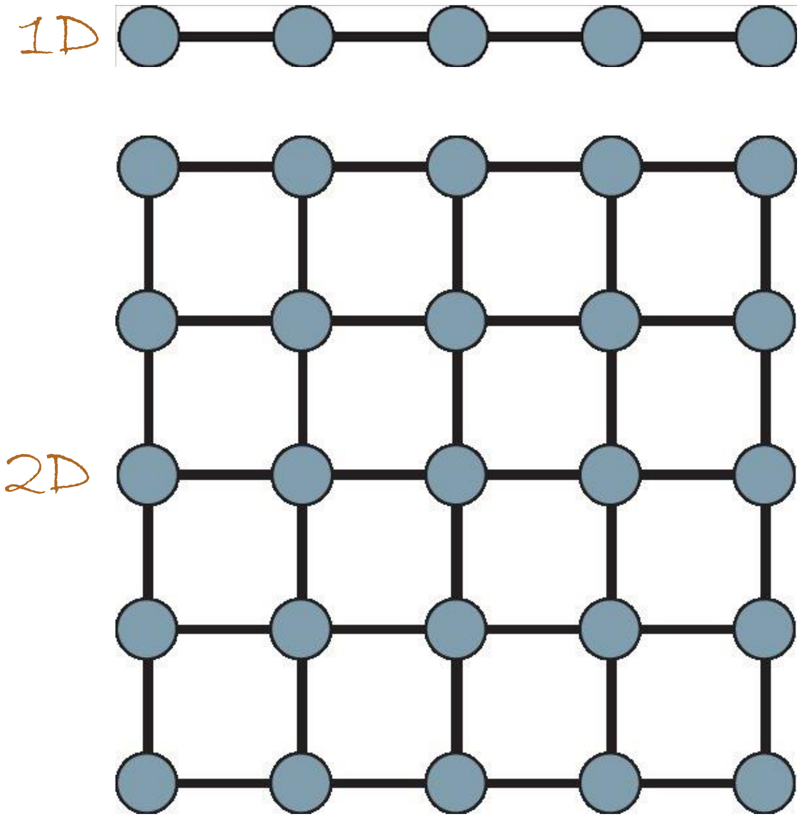
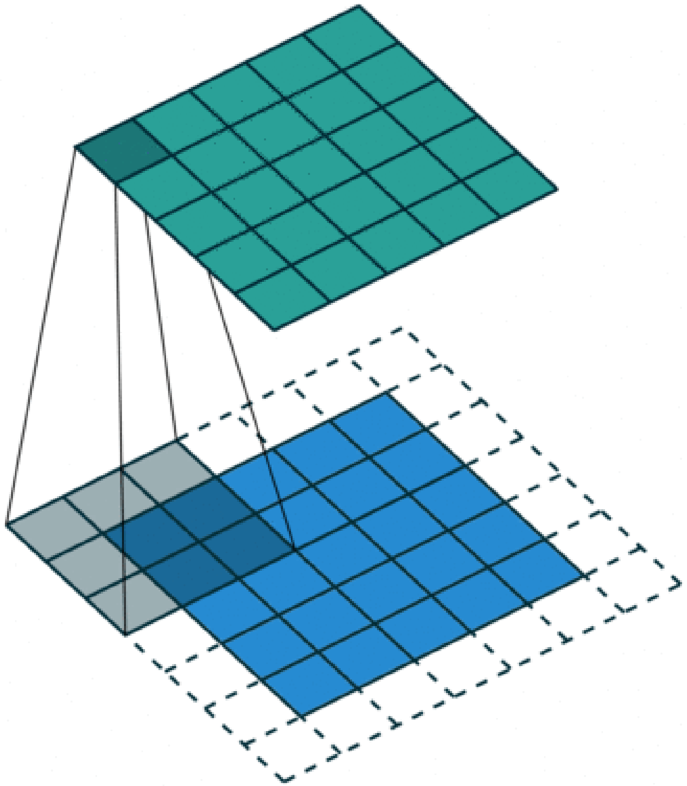
Convolutional Neural Networks

Convolutional Neural Networks (CNN)

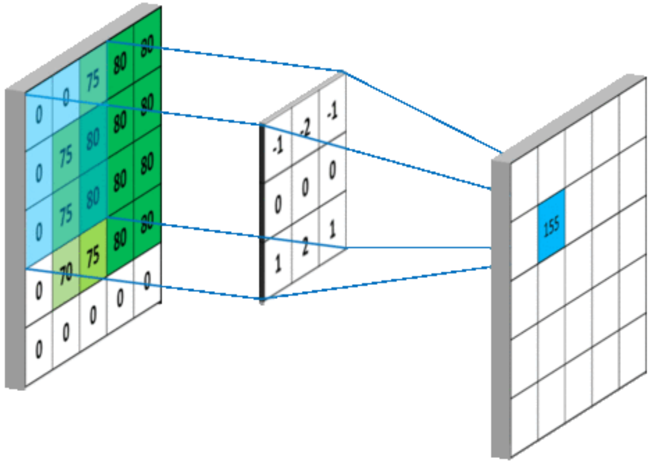
grid-like topology



convolution



pooling



The Convolution Operation

track car location

The Convolution Operation

Weighted averages

kernel

$$s(t) = \int x(a)w(t-a)\mathrm{d}a = (x * w)(t)$$

feature map

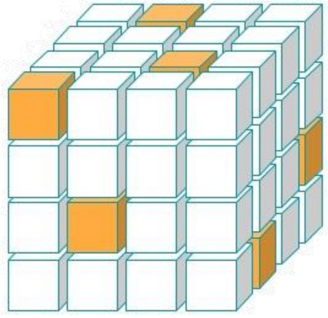
location

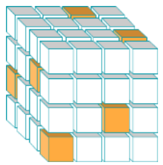
Discrete Convolution

Sampling in a discrete fashion yields

$$s(t) = (x * w)(t) = \sum_{a=-\infty}^{\infty} x(t)w(t-a)$$

Convolution in ML


input

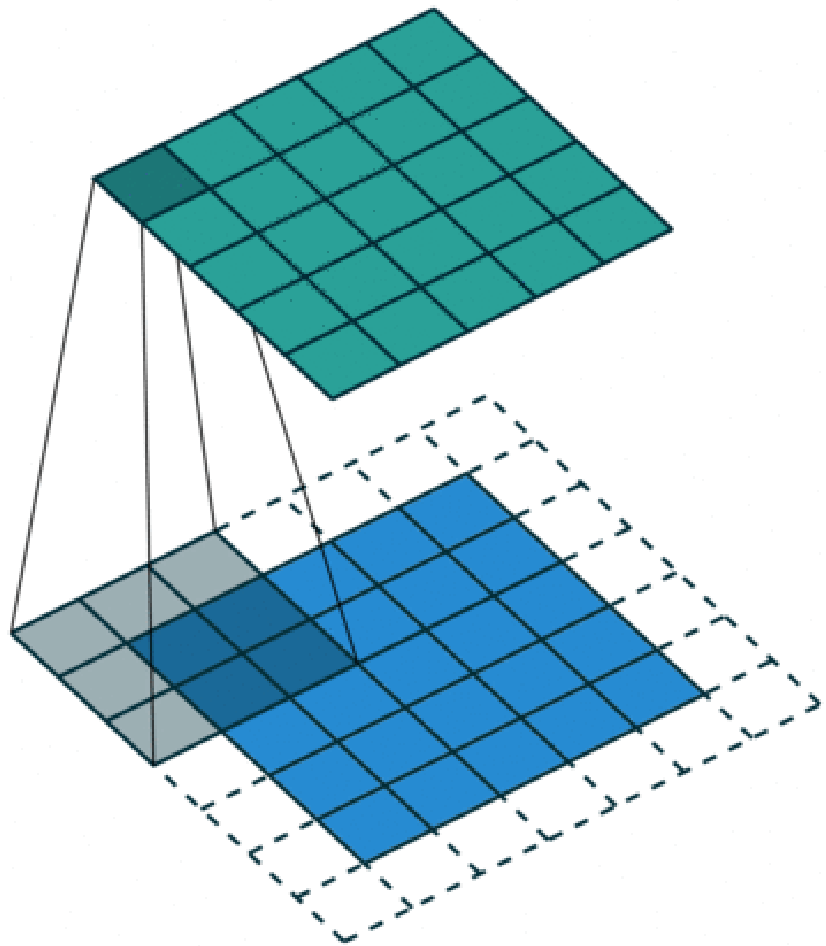

kernel

The multidimensional convolution is given by

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n)K(i-m, j-n)$$

input kernel

Illustrating Convolutions



Cross Correlation vs. Convolution

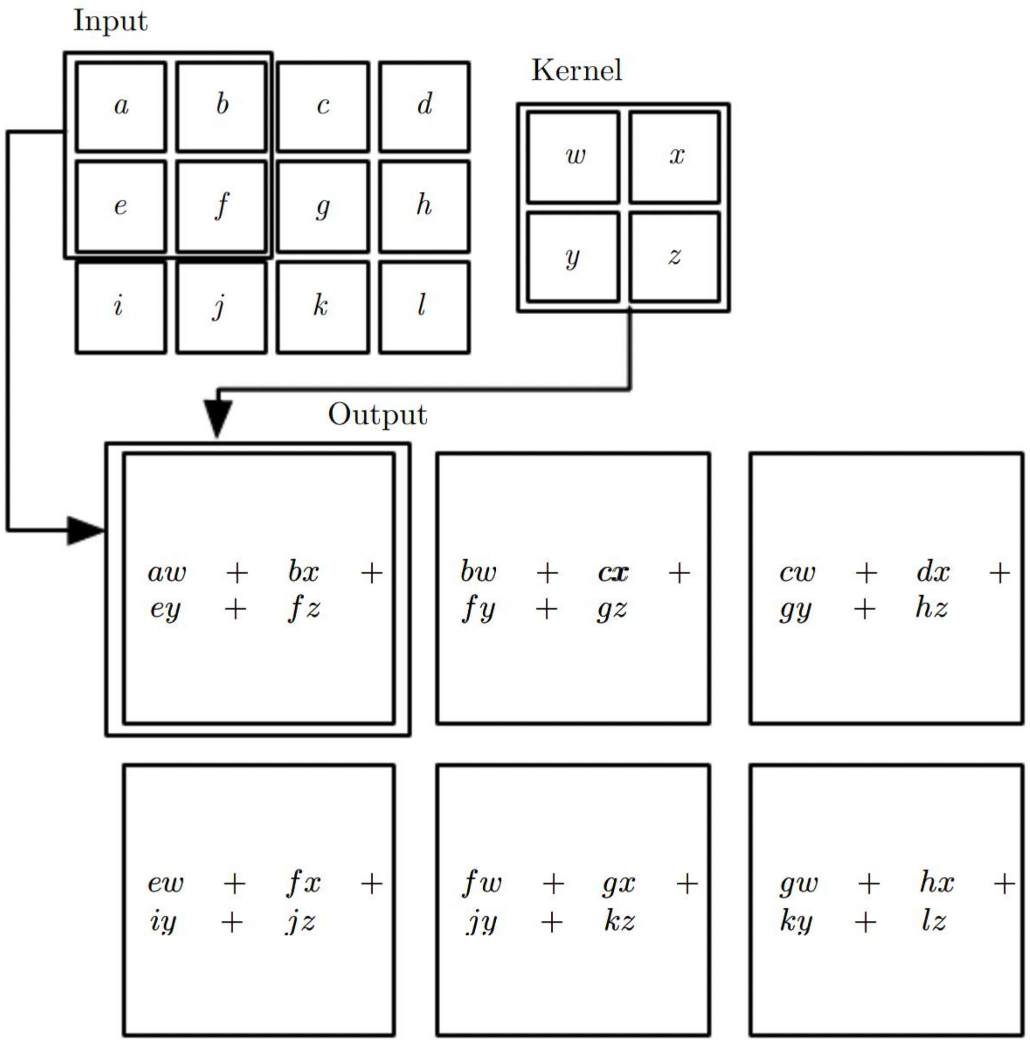
ML (flipped) convolution

$$(I * K)(i,j) = \sum_m \sum_n I(i-m, j-n)K(m,n)$$

ML cross correlation

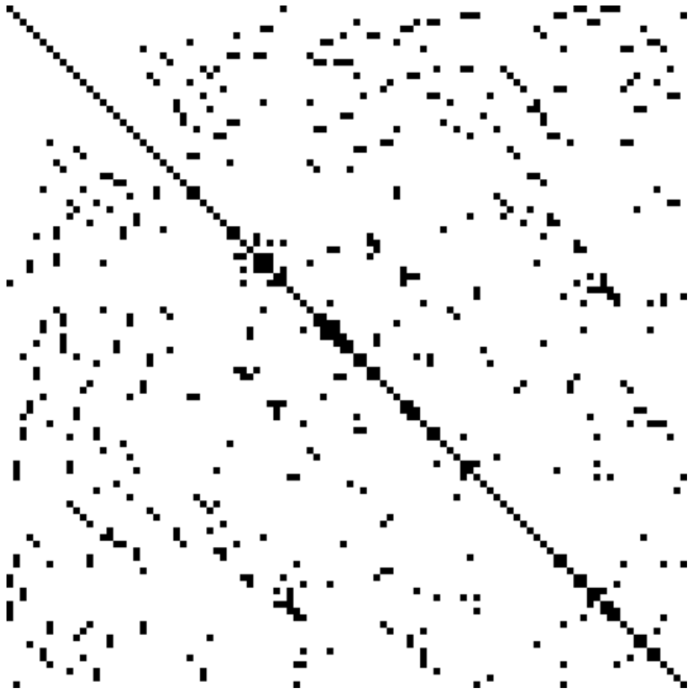
$$(I * K)(i,j) = \sum_m \sum_n I(i+m, i+n)K(m,n)$$

Example: 2D convolution

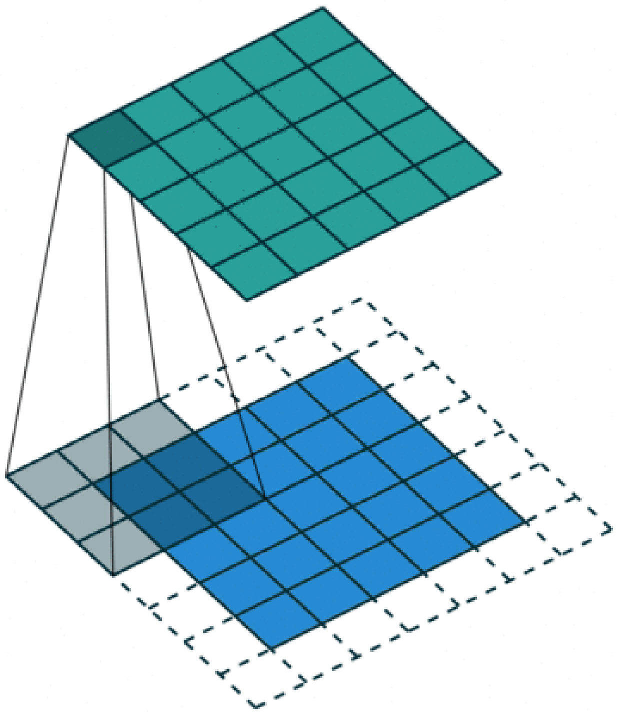


Motivation for CNN

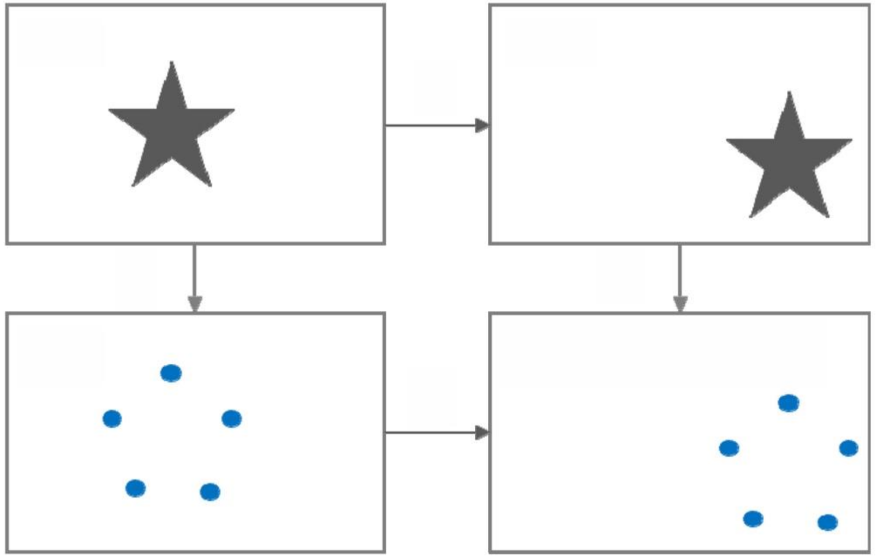
Sparse interactions



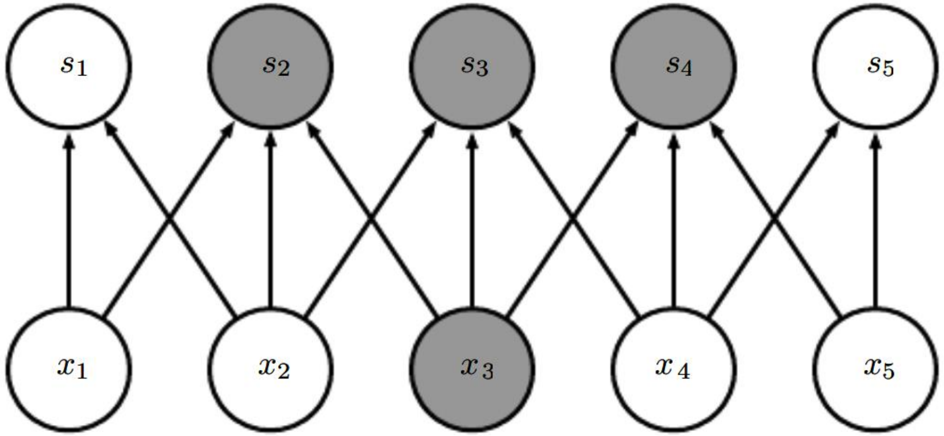
weight sharing



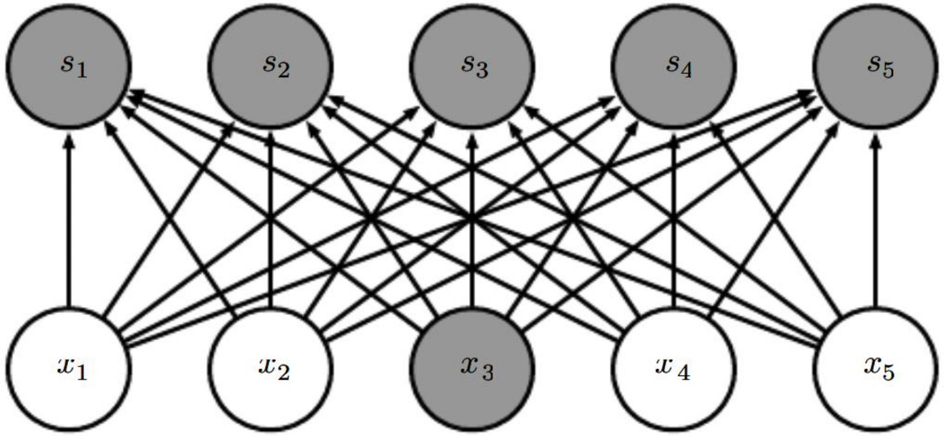
equivariant representation



Sparse Connectivity

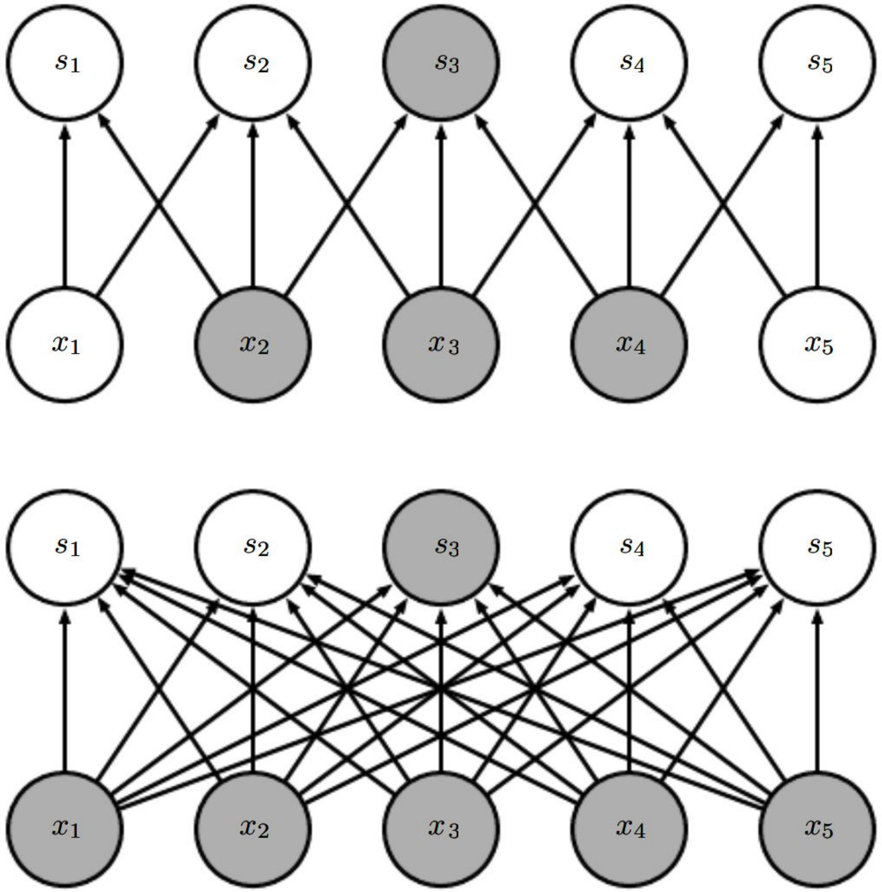


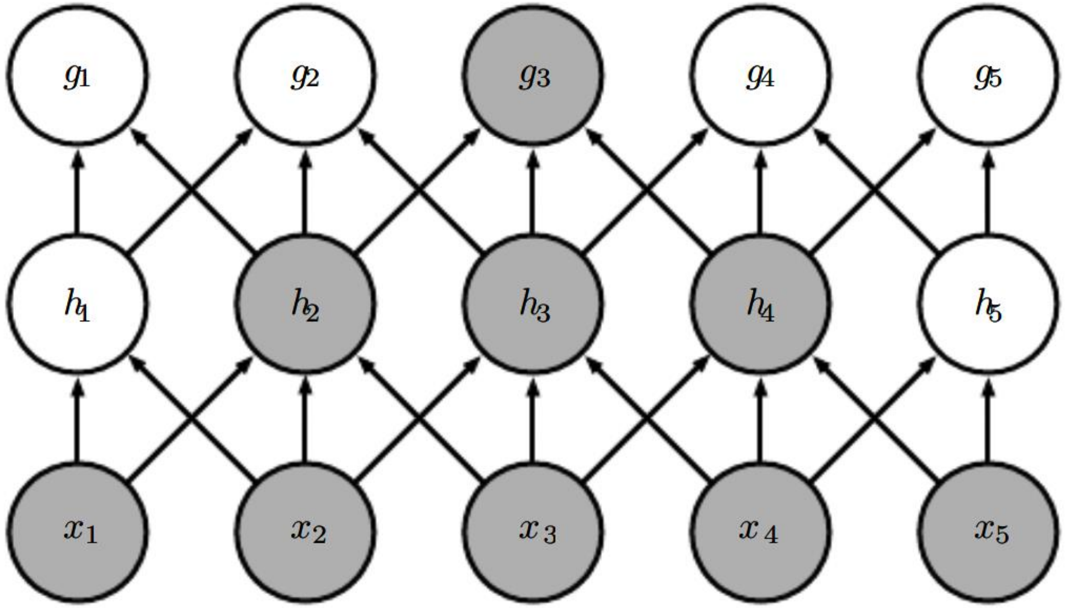
memory + runtime
$O(kn)$



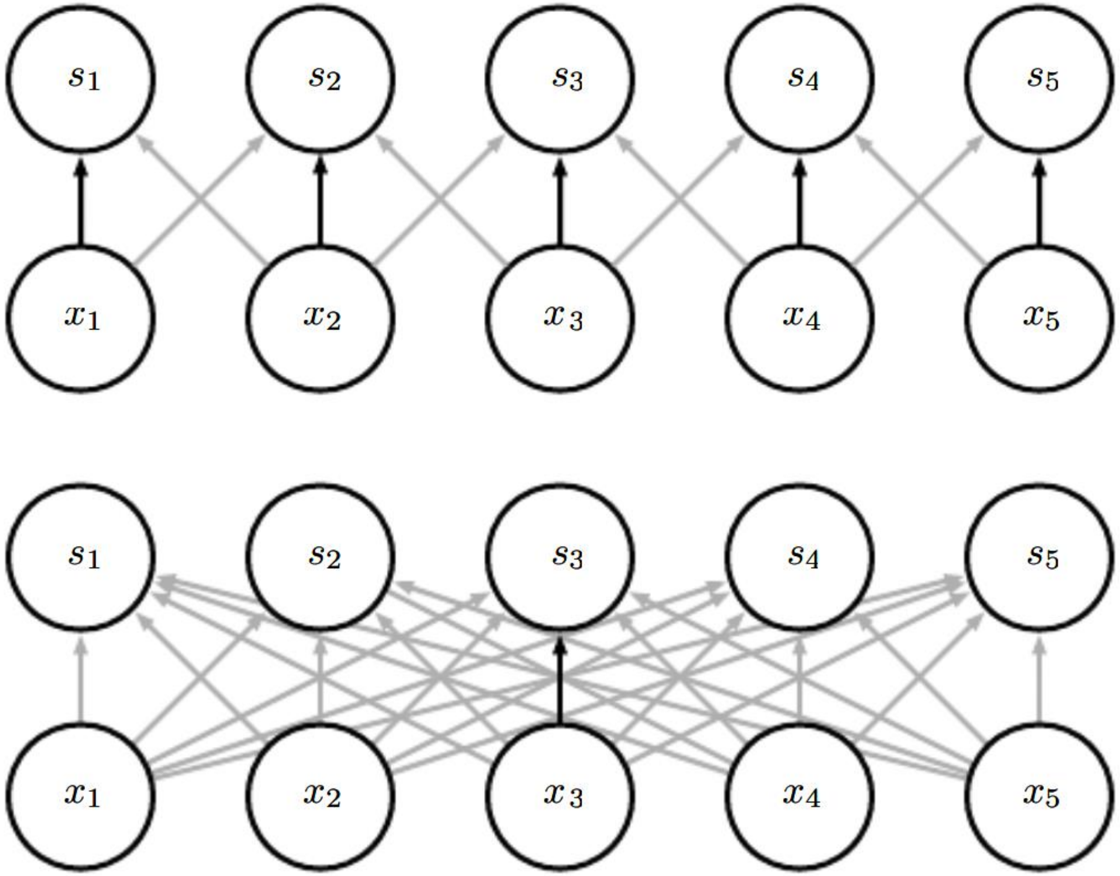
memory + runtime
$O(mn)$

Sparse Connectivity: Receptive Field

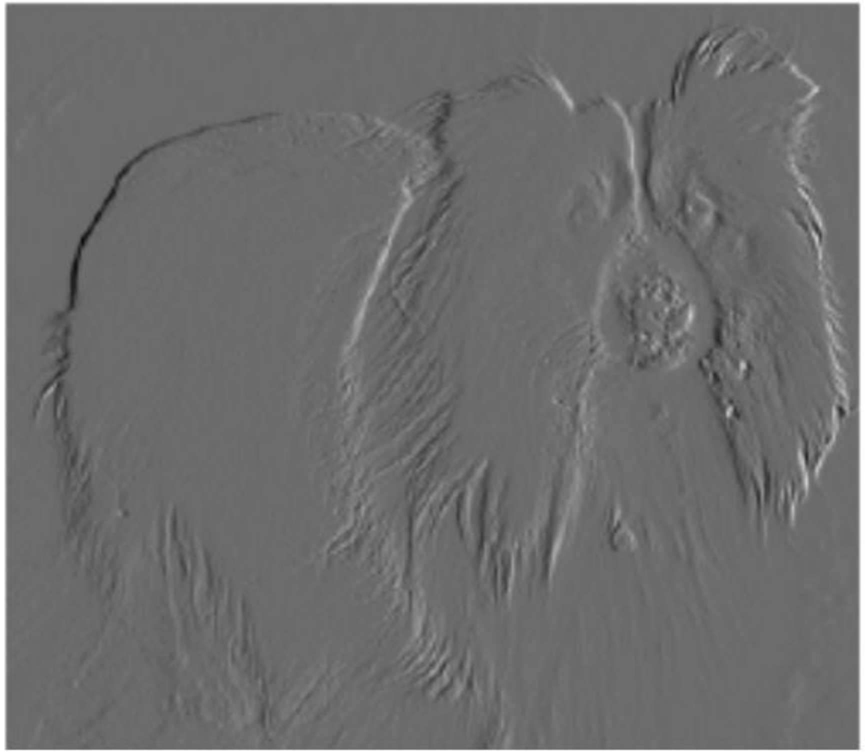




Weight Sharing



Effectiveness of Sparsity & Weight Sharing



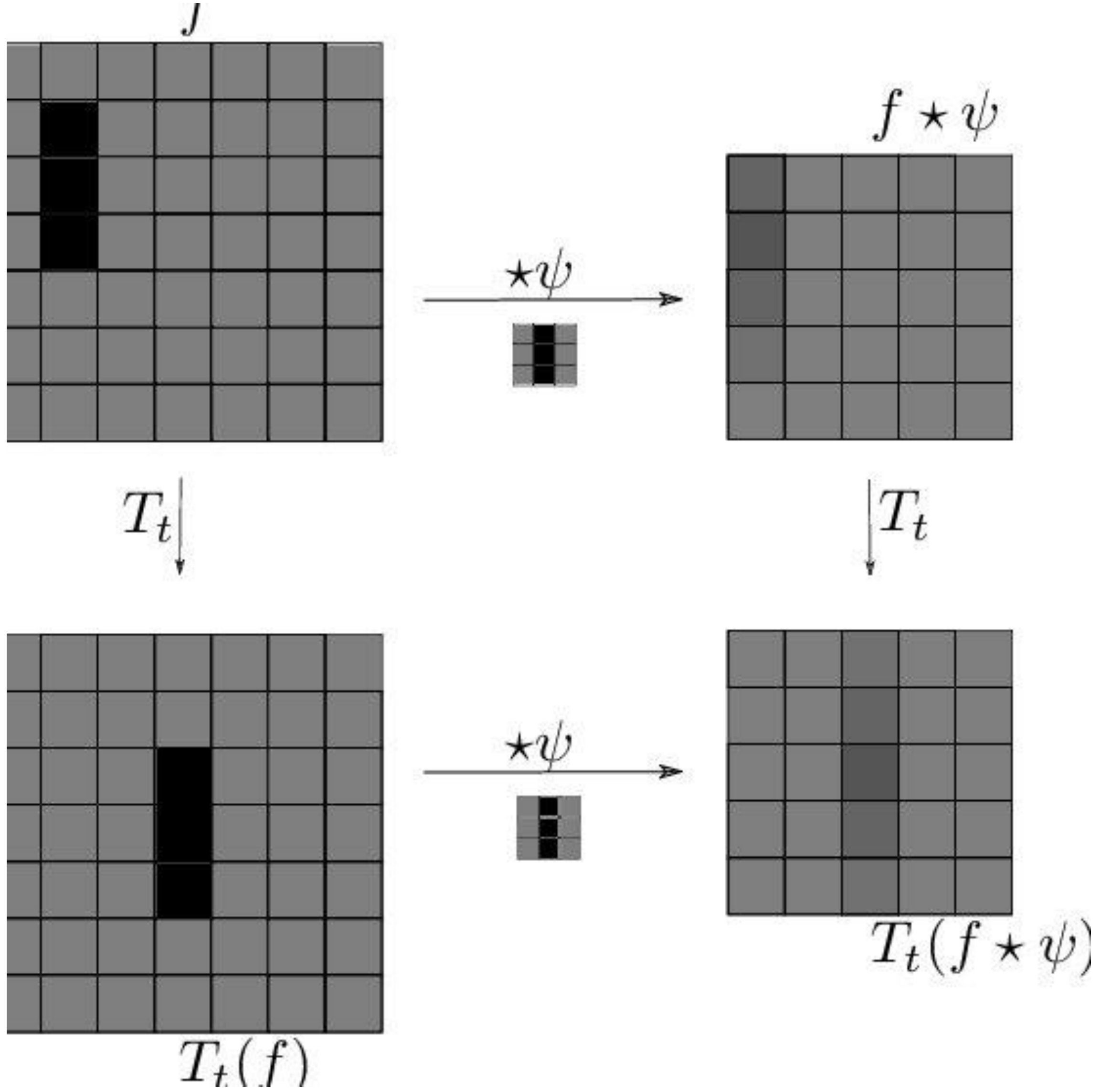
Equivariance to Translations

Weight sharing leads to equivariance to translations

$$g(f(x)) = f(g(x))$$

translation convolution

Equivariance to Translations



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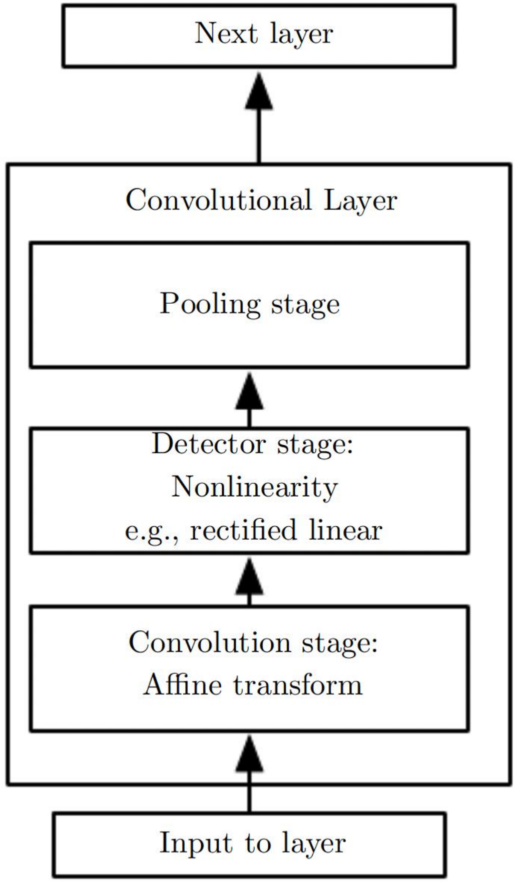
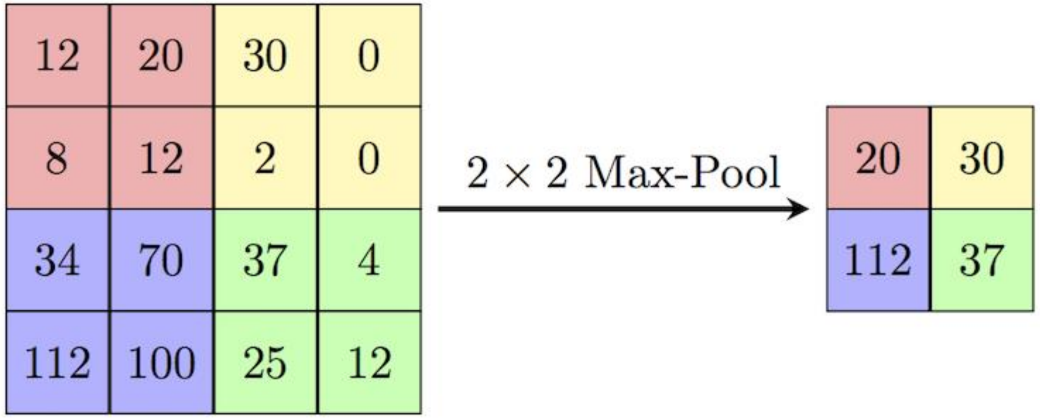
Pooling

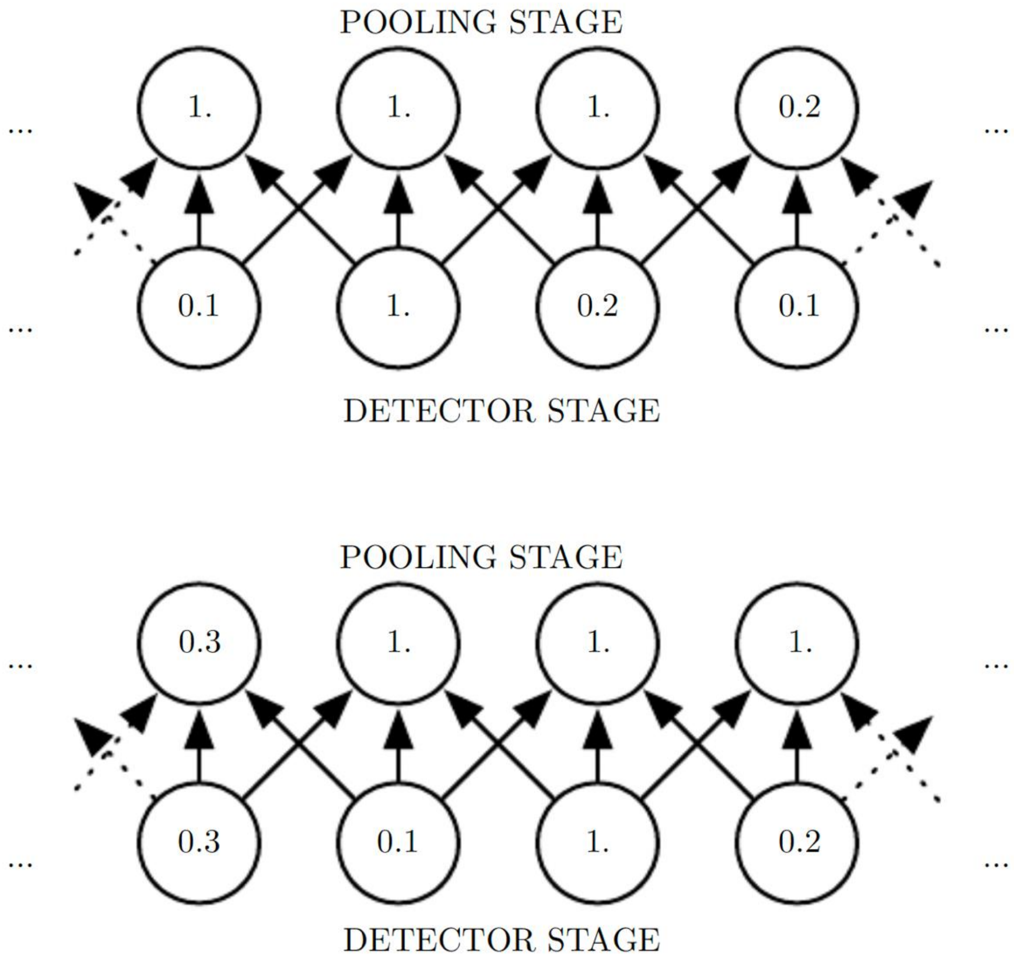


Replace features with summary

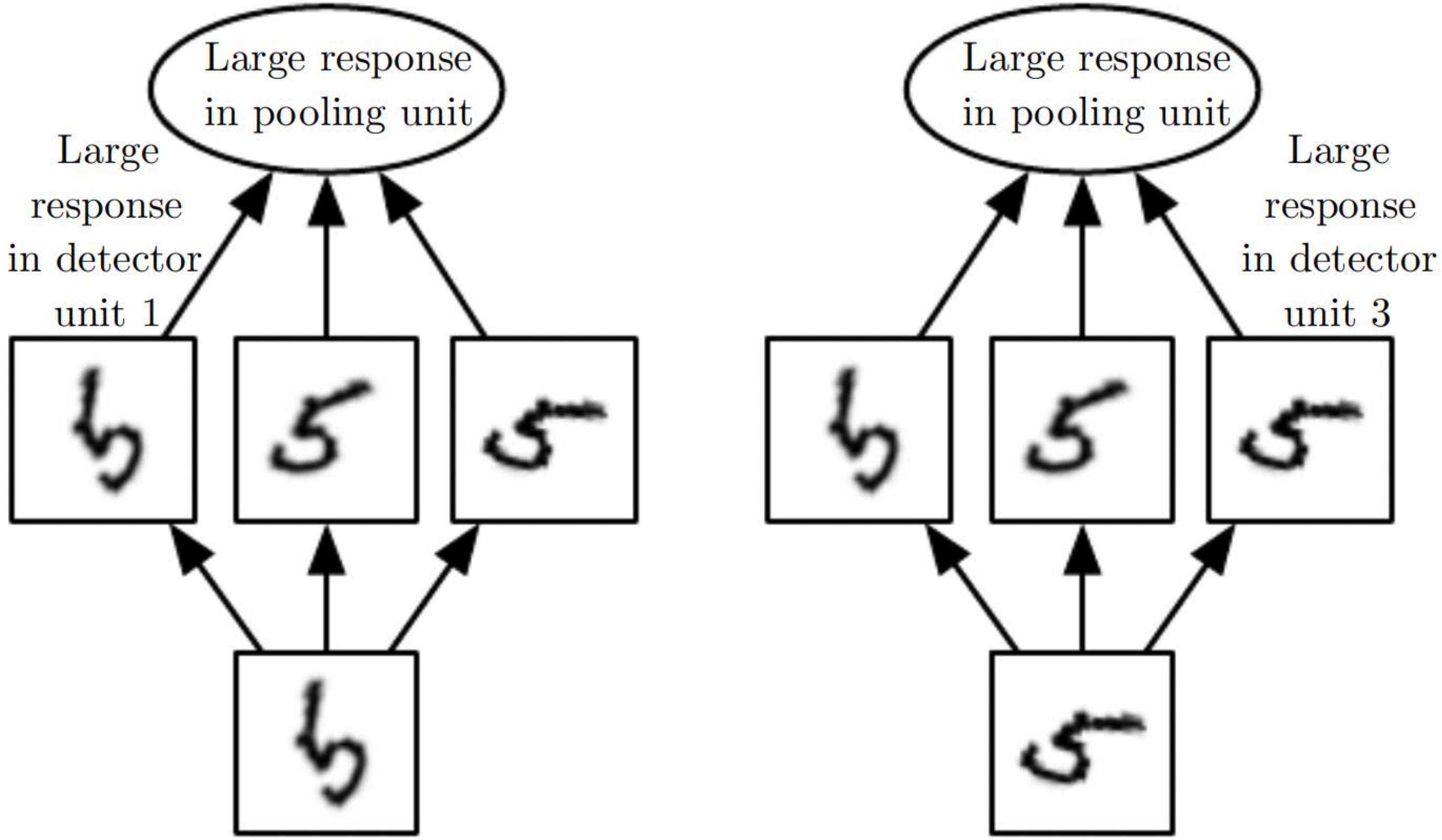
max pooling



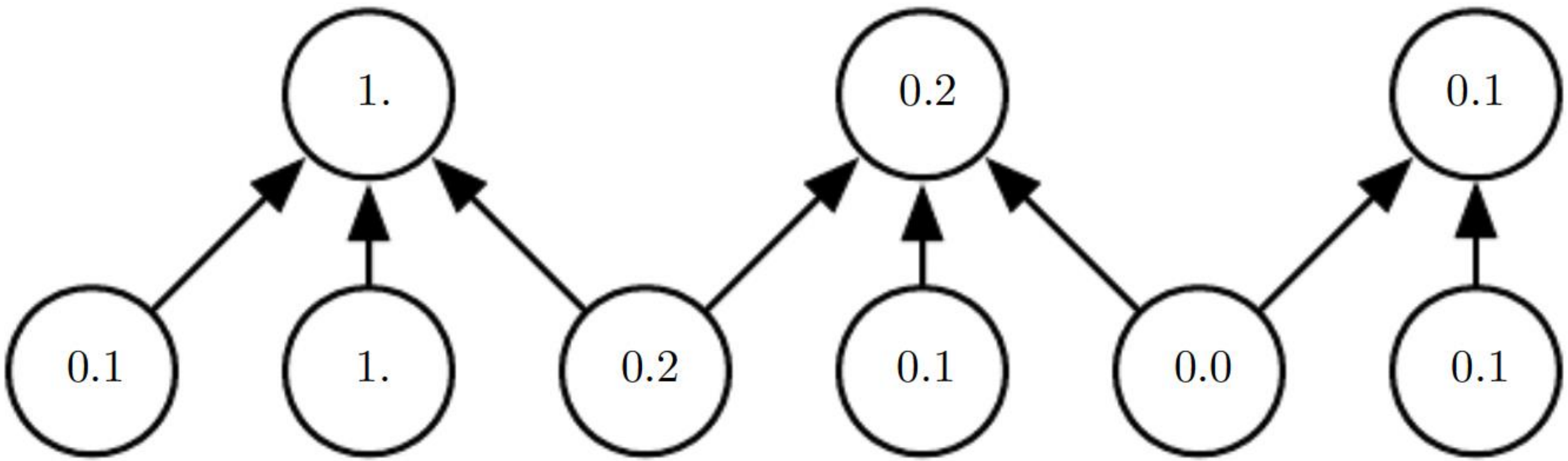
Pooling $\Longrightarrow$ Invariance to Small Translations



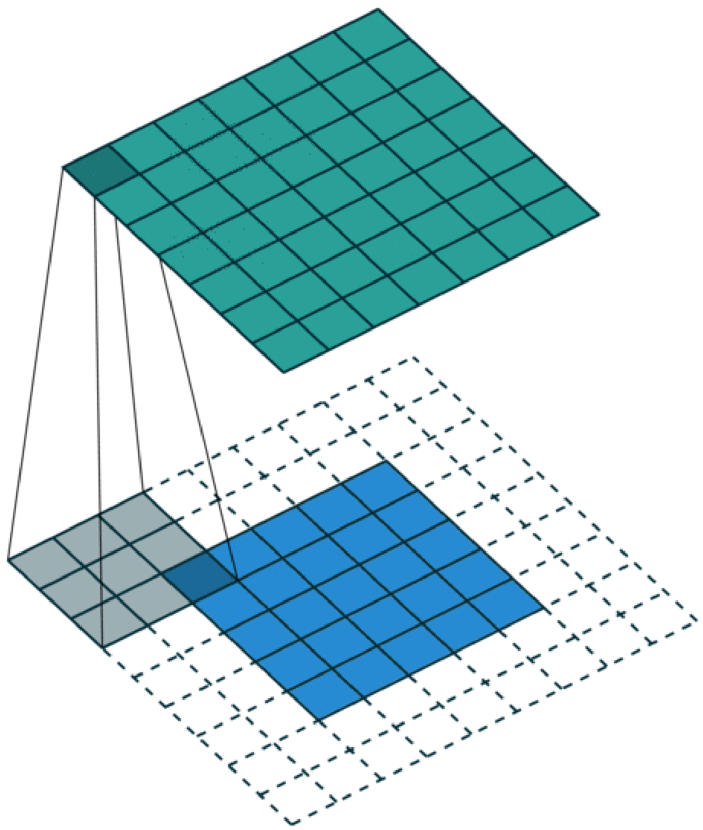
Learned Invariances

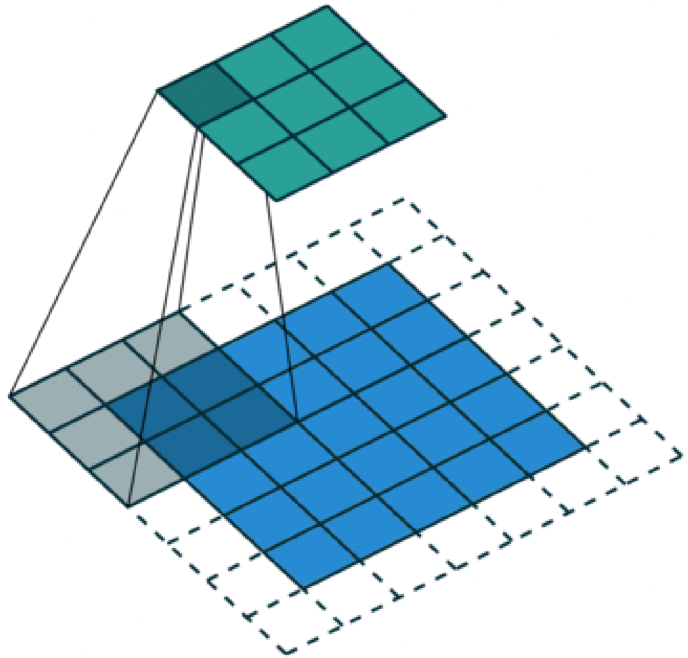


Pooling with Stride> 1



Convolution with Stride





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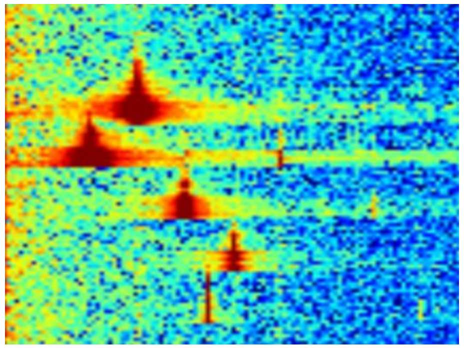
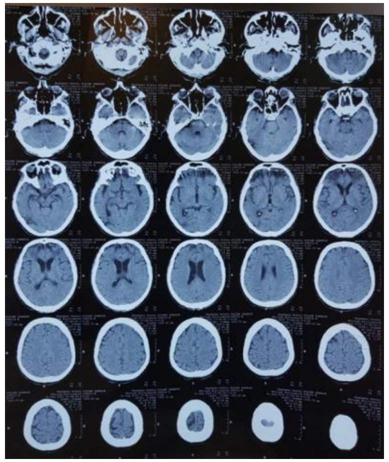
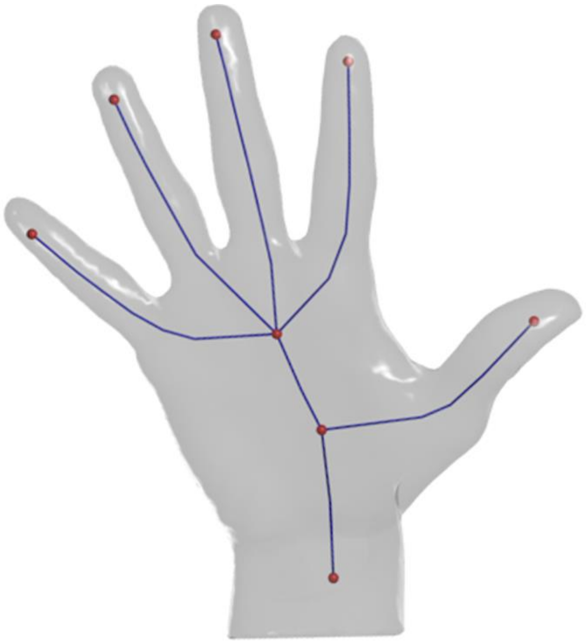
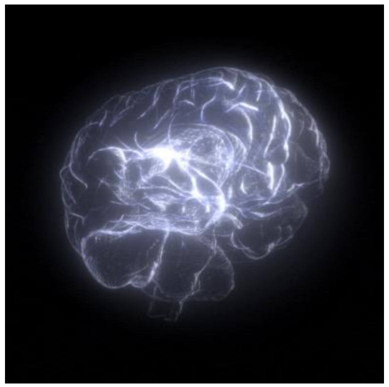
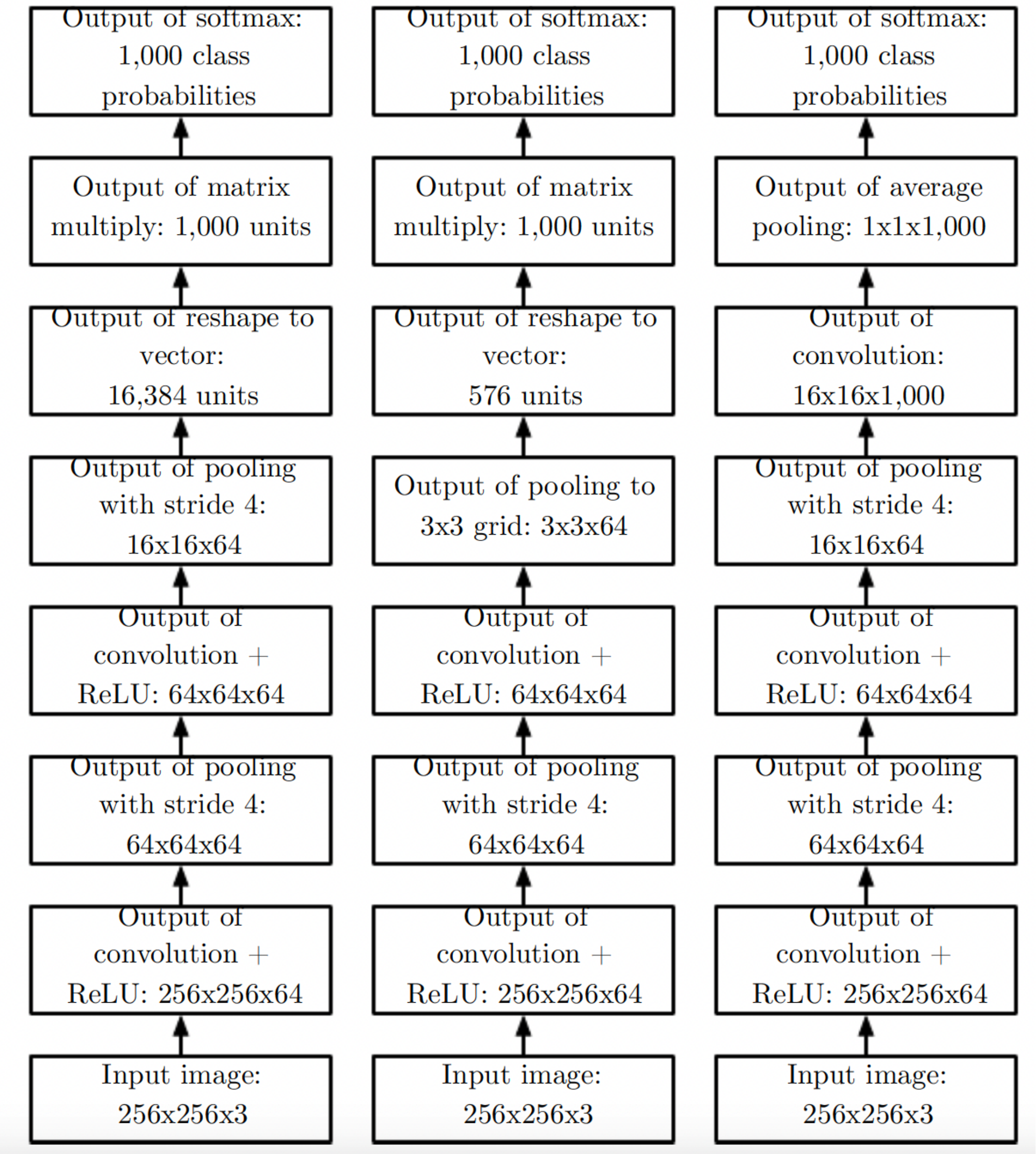
Data Types

	1D	2D	3D
single channel			
multi channel			

Variable Size Inputs

same layers



Code Demo